

Analog Output MAPS Template (Instrumentation)

The Analog Output MAPS template is designed to control an analog that is connected to an analog control unit.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in three models – advanced, standard and basic, which provide respectively lower levels of control over the analog output.

This document describes the Advanced Analog Output MAPS Template and also defines the level of functionality provided by the standard and basic models. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the analog being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Note: The version 2 (v2) Analog Output MAPS template, communicates Floating point values instead of Integer values to the driver. The High-High and Low-Low Alarms does not latch as does not set the FAULT active. The Fault Active is only when the signal is out of range, meaning over or under range.

Advanced Analog Output Functional Philosophy

The analog output is scaled and alarmed according to the setpoints from the Setup faceplate.

Field Outputs

- Analog Output

PLC interlocks

- High-high Alarm
- High Alarm
- Low Alarm
- Low-low Alarm

Simulation mode - will disable the physical analog output and simulate an analog in the PLC program that is controlled from the faceplate. Simulation mode is used to test the process logic without receiving the physical output itself. It is very useful in testing, commissioning and maintenance.

Typical Faceplate Graphics for the Advanced Analog Output MAPS Template

Analog Output					
					
0000_ABCDE_123		Control/Tag name			
123456		Analog Indication Value			
		Fault active			
		Manual mode			
		High-High and Low-Low Alarm Active (Version 2 only)			
		High and Low Alarm Active (Version 2 only)			
Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshal	x	x	x
SSCL 1	High-high Alarm	Analog	x	x	
SSCL 2	High Alarm	Analog	x	x	
SSCL 3	Low Alarm	Analog	x	x	
SSCL 4	Low-low Alarm	Analog	x	x	
SSCL 5	Scale Min In Setpoint (Raw Min)	Analog	x		
SSCL 6	Scale Max In Setpoint (Raw Max)	Analog	x		
SSCL 7	Scale Min Out Setpoint (Eng. Min)	Analog	x		
SSCL 8	Scale Max Out Setpoint (Eng. Max)	Analog	x		
SSCL 9	Simulation Value	Analog	x	x	x
SSCL 10	Alarm Hysteresis	Analog	x	x	

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshal	x	x	x
SSSL 1	Analog Output Process Value (Scaled)	Analog	x	x	x
SSSL 2	Analog Output RAW Value	Analog	x		

Analog Output:				
AO 0	Analog Output for Analog Output	x	x	x

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Simulation	x	x	x	x
Bit 1	High-high AO Alarm Enabled	x	x	x	
Bit 2	High AO Alarm Enabled	x	x	x	
Bit 3	Low AO Alarm Enabled	x	x	x	
Bit 4	Low Low AO Alarm Enabled	x	x	x	
Bit 5	Reset Fault		x	x	x
Bit 6	Spare				
Bit 7	Spare				
Bit 8	Spare				
Bit 9	Spare				

Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Simulation On		x	x	x
Bit 1	High-high Alarm Active	x	x	x	
Bit 2	High Alarm Active	x	x	x	
Bit 3	Low Alarm Active	x	x	x	
Bit 4	Low-low Alarm Active	x	x	x	
Bit 5	Warning Active	x	x	x	
Bit 6	Alarm Active	x	x	x	
Bit 7	Alarm Fault Latch		x	x	x
Bit 8	Alarm Setup Fault		x	x	
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

Resources	Advanced	Standard	Basic
Digital Inputs	0	0	0
Digital Outputs	0	0	0
Analog Inputs	0	0	0
Analog Outputs	1	1	1
PLC Memory Steps	355	271	82
System Words Used	40	40	2
System Bits Used	71	63	51
SCADA SCAN Points	14	9	4

Template: Advanced Analog Output PLC Function Block

<Device> (Advanced Analog Output Control)

	FB_AO_A_V1_0		
	AO_A_v1_0		
DATA_AO_A_V1_0.VAR_Auto_SP	VAR_AO_AutoSP	F_O_AO_MV_RAW_PV	DATA_AO_A_V1_0.F_O_AO_Raw
DATA_AO_A_V1_0.S_CW1_Marshal	S_CW1_Marshal	S_SW1_Marshal	DATA_AO_A_V1_0.S_SW1_Marshal
DATA_AO_A_V1_0.S_CW2_HH_Alarm	S_CW2_HH_Alarm	S_SW2_AO_PV	DATA_AO_A_V1_0.S_SW2_AO_PV
DATA_AO_A_V1_0.S_CW3_H_Alarm	S_CW3_H_Alarm	S_SW3_AO_Raw_PV	DATA_AO_A_V1_0.S_SW3_AO_Raw_PV
DATA_AO_A_V1_0.S_CW4_L_Alarm	S_CW4_L_Alarm	S_SW1_0_Manual	
DATA_AO_A_V1_0.S_CW5_LL_Alarm	S_CW5_LL_Alarm	S_SW1_1_HH_Alarm	
DATA_AO_A_V1_0.S_CW6_MinIn	S_CW6_MinIn	S_SW1_2_H_Alarm	
DATA_AO_A_V1_0.S_CW7_MaxIn	S_CW7_MaxIn	S_SW1_3_L_Alarm	
DATA_AO_A_V1_0.S_CW8_MinOut	S_CW8_MinOut	S_SW1_4_LL_Alarm	
DATA_AO_A_V1_0.S_CW9_MaxOut	S_CW9_MaxOut	S_SW1_5_Over_Range	
DATA_AO_A_V1_0.S_CW10_Manual_SP	S_CW10_Manual_SP	S_SW1_6_Under_Range	
DATA_AO_A_V1_0.S_CW11_Alarm_Hysteresis	S_CW11_Alarm_Hysteresis	S_SW1_7_Fault	
		S_SW1_8_Setup_Fault	

Template Graphic: Advanced Home Screen

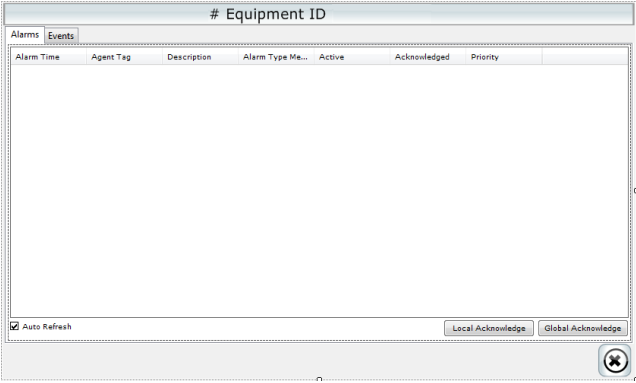
The graphic shows a control interface for an analog output. It includes a 'Status' section with a numeric display (123456), a red alarm icon, and buttons for 'Over range', 'Under range', 'Error', and 'Reset'. A 'Control' section features a green level bar and 'Limits' (High High, High, Low, Low Low) with corresponding red indicator lights. A 'Manual' section has a 'Manual' button and a numeric display (123456). At the bottom are icons for a wrench, a warning triangle, and a crossed-out circle. An arrow points from the 'Manual' button to the 'Manual' entry in the adjacent table.

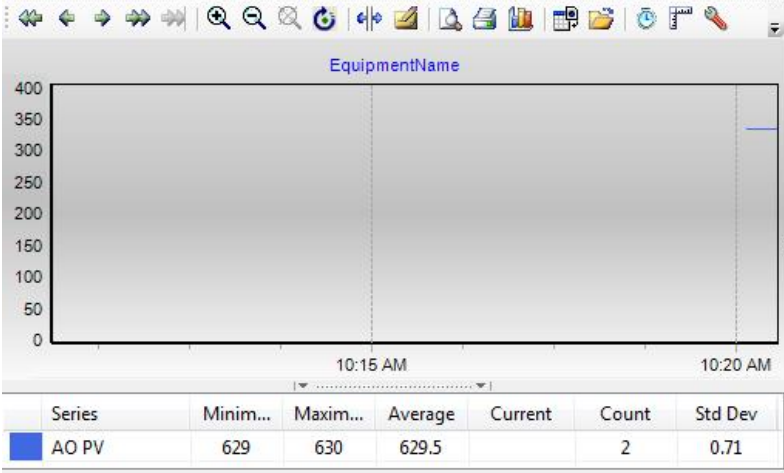
Control	Descriptions
# Equipment ID	Analog Output controlled name
High High	High-high alarm indication
High	High alarm indication
Low	Low alarm indication
Low Low	Low-low alarm indication
Over range	Analog over range alarm
Under range	Analog under range alarm
Error	Alarm fault latch
Reset	Reset the analog Output after a fault has been repaired
123456	Analog scaled value indication
Manual	Manual mode enable button
123456	Manual mode analog output default value
H	Manual indication
	Alarm fault latch

Template Graphic: Advanced Setup Screen

The graphic shows a setup interface for an analog output. It includes 'Analog Eng In' and 'Analog Raw Out' displays (both 123456). A 'Scaling' section has 'Eng' and 'Raw' columns with 'Min' and 'Max' input fields. A 'Hysteresis' section has an 'Alarm Hysteresis' field (#####) and a 'Unit' field (#####). An 'Alarm Limits' section has 'High High', 'High', 'Low', and 'Low Low' fields, each with a value (123456) and an 'Enable Tick' checkbox. A 'Warning: Possible Setup Fault' message is displayed. At the bottom is a crossed-out circle icon.

Control	Descriptions
Analog Raw Out	Raw value scaled from input value for analog output
Analog Eng In	Value from PLC for analog
Min, Max Raw	Raw output values for scaling
Min, Max Scaled	Engineering input values for scaling
High High	High-high alarm setup value and enabling
High	High warning setup value and enabling
Low	Low warning setup value and enabling
Low Low	Low-low alarm setup value and enabling
Alarm Hysteresis	High and low warning auto reset hysteresis value. This is a value at which the warnings will reset should the process value decrease – in the case of the high level or increase – in the case of a low level
Unit	Unit displayed for the scaled analog

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Equipment Name	Display the name of the current analog Output of the trend
	AO PV	Display the level of the current level